

13.3 Applying Properties of Operations

Lesson Introduction

 Benchmark	MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.		 Learning Target	I can apply the associative, commutative, and distributive properties to generate equivalent expressions.
 Benchmark Coherence	Building On MA.5.AR.2 Demonstrate an understanding of equality, the order of operations, and equivalent numerical expressions.		Working Toward MA.7.AR.1.2 Determine whether two linear expressions are equivalent.	
 Essential Questions	- Which operations does each property (associative, commutative, distributive) apply to? - How can you use the properties of operations to write equivalent expressions? - How can you show that two algebraic expressions are equivalent?			
 Lesson Narrative	This lesson begins by revisiting the associative, commutative, and distributive properties that students are familiar with from prior grades. The lesson primarily focuses on students exploring these properties for numerical expressions to ensure students recall the properties clearly. Students' connections of these properties to algebraic expressions is scaffolded throughout the lesson.			
 Component Summary	13.3.1 Warm-Up (~5 min.)	Rewrite expressions using the distributive property of multiplication over addition (<i>SE p. 274</i>)	13.3.3 Guided Instruction (20-25 min.)	Apply the properties of operations to write equivalent expressions (<i>SE p. 276</i>)
	13.3.2 Exploration (15 min.)	Apply the properties of operations to write equivalent expressions to represent the areas of rectangles (<i>SE p. 275</i>)	13.3.4 Wrap-Up (~5 min.)	Demonstrate understanding of using the properties of operations to write equivalent expressions (<i>SE p. 279</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Equivalent - Commutative Property of Multiplication - Commutative Property of Addition - Associative Property of Multiplication - Associative Property of Addition <u>Additional Terms:</u> - Distributive property		(Optional) Colored pencils	Prepare an exemplar math journal or interactive notebook entries to show students your expectations.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle to recognize that expressions like $-4x + 3y$ and $3y - 4x$ are equivalent expressions by the commutative property. - Students may not realize that the number of elements inside of the parentheses is not limited to two elements when applying the distributive property.	Consider using word associations to help students understand the properties of operations. For example, relating commutative property to commuting, as in moving from one location to another. Similarly, the teacher can relate associative property to associates or associating with people in groups.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns In 13.3.3 Guided Instruction, students can take turns explaining, justifying, agreeing and disagreeing, or asking clarifying questions as they complete the tables of equivalent expression. Students' understanding is deepened as they construct logical arguments and critique the reasoning of others.	MA.K12.MTR.1.1: Participate in Effortful Learning Throughout 13.3.3 Guided Instruction, students collaborate to extend their understanding of the properties of operations. Students work together to support each other as they build their understanding.
 Differentiate Instruction	Consider Frayer Models, interactive notebooks, word walls, and other vocabulary strategies to assist students in their language acquisition development with the properties of addition and multiplication.	Definition Characteristics Examples Non-examples Property
Lesson Notes:		

13.3.1 Warm-Up: Mental Math

Component Overview: In this Warm-Up, students have a chance to connect the distributive property to mental mathematics.



13.3.1 Warm-Up: Mental Math

Ulrich finds the product of $5 \cdot 107$ in his head. Ulrich wrote the following explanation of his thinking:

Because 107 is equal to $100 + 7$, I can write $5 \cdot 107$ as $5(100 + 7)$. I know that $5 \cdot 100$ is 500 and that $5 \cdot 7$ is 35. Then, I add 500 and 35 to get 535.

1. Use Ulrich's reasoning to write each in the form $a(b + c)$. Then, use that form to find the product.

Multiplication	$5 \cdot 104$	$5 \cdot 97$	$5 \cdot 130$
$a(b + c)$	$5(100 + 4)$	$5(90 + 7)$	$5(100 + 30)$
Product	$500 + 20 = 520$	$450 + 35 = 485$	$500 + 150 = 650$



Consider pairing students with different strengths for this activity and providing space for them to compare strategies for collective learning.



- How can you rewrite 104, 97, and 130 using the method that Ulrich used?
- How does using 100 make multiplying the two numbers easier to do using mental math?

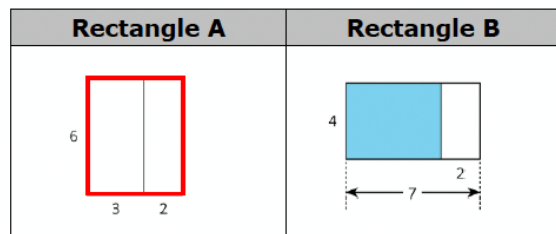
13.3.2 Exploration: Representing Area of a Rectangle

Component Overview: The 13.3.2 Exploration uses the area model to help students visualize the distributive property. The exploration also leads students to seeing equivalence in numeric expressions, then connecting that understanding to be applied to algebraic expressions.



13.3.2 Exploration: Representing Area of a Rectangle

1. Work with your partner to answer the questions.



This component provides a visual entry point by connecting the area model to the distributive property. Highlight this connection and encourage students to regularly use the area model.

- a. Rectangle A is divided into two smaller rectangles. Select all the expressions that could be used to determine the area of Rectangle A.

- ☐ $6 \cdot 5$
☐ $6 \cdot 3 \cdot 2$
☒ $6(3 + 2)$
☐ $6 \cdot 3 + 2$
☐ $6 + 3 + 2$
☒ $6 \cdot 3 + 6 \cdot 2$

- b. With your partner, discuss which expressions are in the form $a(b + c)$.
 $6(3 + 2)$

- c. Rectangle B is divided into two smaller rectangles. Select all the expressions that could be used to determine the area of the blue rectangle inside Rectangle B.

- ☒ $4 \cdot 5$
☐ $4 \cdot 7 \cdot 2$
☒ $4(7 + 2)$
☒ $4 \cdot 7 + 4 \cdot 2$
☐ $4 \cdot 7 + 4 \cdot 2$
☐ $4 \cdot 2 - 4 \cdot 7$

13.3.2 Exploration: Representing Area of a Rectangle (continued)

- d. With your partner, discuss which expressions are in the form $a(b + c)$.

$$4(7 - 2) = 4(7 + (-2))$$

2. Complete the following problems.

- a. Circle all of the expressions that have the same value as $-4(x + 7)$ if $x = -13$.

$-4x + 7$	$(x + 7) \cdot -4$	$-4x + -4 \cdot 7$	$4(x - 7)$	$-28 - 4x$
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- b. With your partner, discuss whether or not the expressions you circled would change if the x -value were a different number. Summarize your decision and reason.

The expressions are equivalent for any value of x .



- What do you already know about substituting variables in algebraic expressions?
- Why do you think the three expressions circled resulted in the same value?
- Why do you think that the same expressions will be circled for any value of x ?

13.3.3 Guided Instruction: Applying Properties of Operations

Component Overview: Students formally define the distributive, commutative, and associative properties. Students practice applying each property to evaluate numeric and algebraic expressions.



13.3.3 Guided Instruction: Applying Properties of Operations

The **distributive property of multiplication over addition** is a property that applies to all rational numbers. The distributive property “distributes” a factor to all terms within a set of parentheses.

- Complete the table. The first row has been completed.

$a(b + c)$	$ab + ac$	Value
$-8(50 + 9)$	$-8(50) + -8(9)$	-472
$17(20 + 3)$	$17(20) + 17(3)$	$340 + 51 = 391$
$(100 - 7) \cdot -6$	$100 \cdot -6 + -7 \cdot -6$	$-600 + 42 = -558$
$(40 + 72)29$	$40 \cdot 29 + 29 \cdot 72$	$1160 + 2088 = 3248$
$-11(48 - 96)$	$-11(48) - -11(96)$	$-528 + 1056 = 528$

- In two of the rows the expression had a subtraction sign instead of an addition sign. Discuss with your partner how $a(b - c)$ can be rewritten using addition. Write a summary of your discussion.
 $a(b + (-c))$
- Use the expressions below to show that the distributive property of multiplication over addition works with rational numbers.

$a(b + c)$	$ab + ac$	Equivalent?
$\frac{1}{2}(-14 + 50)$	$\frac{1}{2}(-14) + \frac{1}{2}(50)$	Yes
$1.2(12 + 50)$	$1.2 \cdot 12 + 1.2 \cdot 50$	Yes

Mathematicians determine a property exists when expressions are equivalent, no matter what number is used. A property can then be used with variables.



Consider your students when planning your approach for 13.3.3 Guided Instruction.

- Teacher-led:** For students unfamiliar with the properties or who struggled with the 13.3.1 Warm-Up, consider modeling each property and providing Frayer Models to complete.
- Small groups:** Nominate student captains that pull small groups and act as co-teachers for the lesson. Alternatively, students can group on their own and report their findings back to the teacher after each property.
- Hybrid:** Utilize instructional routines like Think-Pair-Share or Collect and Display to facilitate guided instruction that positions students to activate what they know about properties and apply that knowledge to new situations within this activity.

13.3.3 Guided Instruction: Applying Properties of Operations (continued)

Expressions are **equivalent** if they have the same value, no matter what number is substituted in for each of the variables.

4. Write an equivalent expression for $8(x + 2)$.

$$8x + 16$$

5. Write an equivalent expression for $\frac{1}{2}(x - 16)$.

$$\frac{1}{2}x - 8 \text{ or } \frac{1}{2}x + (-8)$$

The **commutative property** states that the order in which an operation is performed does not change the value of the expression.

6. Use the pairs of expressions below to determine the operations for which the commutative property is true.

Operation	Example Expressions		Equivalent?
Addition	$\frac{5}{12} + \frac{3}{4}$	$\frac{3}{4} + \frac{5}{12}$	Yes
	$\frac{5}{12} + \frac{9}{12} = \frac{14}{12}$	$\frac{9}{12} + \frac{5}{12} = \frac{14}{12}$	
Subtraction	$\frac{3}{4} - \frac{11}{2}$	$\frac{11}{2} - \frac{3}{4}$	No
	$\frac{3}{4} - \frac{22}{4} = \frac{-19}{4}$	$\frac{22}{4} - \frac{3}{4} = \frac{19}{4}$	
Multiplication	$\frac{5}{6} \cdot \frac{2}{3} = \frac{10}{18}$	$\frac{2}{3} \left(\frac{5}{6}\right) = \frac{10}{18}$	Yes
	$\frac{7}{12} \div \frac{1}{4}$	$\frac{1}{4} \div \frac{7}{12}$	
Division	$\frac{7}{12} \div \frac{1}{4}$	$\frac{1}{4} \div \frac{7}{12}$	No
	$\frac{7}{12} \cdot \frac{4}{1} = \frac{28}{12}$	$\frac{1}{4} \cdot \frac{12}{7} = \frac{12}{28}$	

The commutative property only applies to **addition** and **multiplication**.



Connect other known words to each of the properties.

- Distributive property: fairly distributing objects like candy or money
- Associative property: associating with different people at different times depending on the situation
- Commutative property: commuting from one place to another (switching from home to school)



Applying the properties using rational numbers helps to extend student understanding that these properties do work for all numbers while also supporting students learning of operations with rational numbers. Exploring whether the commutative property works for all four operations helps students understand why there is no commutative property for subtraction and division.

The Associative property is similarly explored.

13.3.3 Guided Instruction: Applying Properties of Operations (continued)

7. Use the commutative property of multiplication to write an expression equivalent to $5 \cdot \frac{3}{20} \cdot 4\frac{4}{5}$.

$$4\frac{4}{5} \cdot 5 \cdot \frac{3}{20} \quad 5 \cdot 4\frac{4}{5} \cdot \frac{3}{20} \quad \frac{3}{20} \cdot 5 \cdot 4\frac{4}{5} \quad \frac{3}{20} \cdot 4\frac{4}{5} \cdot 5 \quad 4\frac{4}{5} \cdot \frac{3}{20} \cdot 5$$

8. Rewrite $x + 9$ using the commutative property.
 $9 + x$
9. Rewrite $(-7)(x)$ using the commutative property.
 $(x)(-7)$



Commutative Property of Addition

$$a + b = b + a$$

Commutative Property of Multiplication

$$a \times b = b \times a$$

The **associative property** states that the way in which numbers are grouped does not change the value of the expression.

10. Use the pairs of expressions below to determine the operations for which the associative property is true.

Operation	Example Expressions	Equivalent?
Addition	$\left(\frac{8}{5} + \frac{3}{10}\right) + \frac{17}{10}$ $\frac{16}{10} + \frac{3}{10} + \frac{17}{10} = \frac{36}{10}$	Yes
Subtraction	$\frac{7}{2} - \left(\frac{3}{4} - \frac{1}{4}\right)$ $\frac{14}{4} - \left(\frac{3}{4} - \frac{1}{4}\right)$ $= \left(\frac{14}{4} - \frac{2}{4}\right)$ $= \frac{12}{4}$	No



Consider an inquiry-based approach to position students as sense makers.

- Do properties ever apply to subtraction or division? Why or why not? Give an example.
- How can you show that two numerical expressions are equivalent?
- How can you show that two algebraic expressions are equivalent?

13.3.3 Guided Instruction: Properties of Operations (continued)

Multiplication	$\frac{2}{3} \cdot \left(\frac{3}{5} \cdot \frac{1}{9} \right)$ $\frac{2}{3} \cdot \frac{3}{45} = \frac{6}{135}$	$\left(\frac{2}{3} \cdot \frac{3}{5} \right) \cdot \frac{1}{9}$ $\left(\frac{6}{15} \right) \cdot \frac{1}{9} = \frac{6}{135}$	Yes
Division	$\left(\frac{1}{5} \div \frac{1}{10} \right) \div \frac{1}{2}$ $\left(\frac{1}{5} \cdot \frac{10}{1} \right) \div \frac{1}{2}$ $= 2 \div \frac{1}{2}$ $= 2 \cdot \frac{2}{1} = 4$	$\frac{1}{5} \div \left(\frac{1}{10} \div \frac{1}{2} \right)$ $\frac{1}{5} \div \left(\frac{1}{10} \cdot \frac{2}{1} \right)$ $= \frac{1}{5} \div \frac{1}{5}$ $= \frac{1}{5} \cdot \frac{5}{1} = 1$	No

The associative property applies only to **addition** and **multiplication**.

- Rewrite $(3.25 + 9.8) + 7.2$ using the associative property.
 $3.25 + (9.8 + 7.2)$
- Rewrite $(x + 15) + -11$ using the associative property.
 $x + (15 + (-11))$
- Rewrite $(-2)(8 \cdot x)$ using the associative property.
 $(-2 \cdot 8) \cdot x$
- Rewrite $6((x + 3) + 2)$ using the associative property.
 $6(x + (3 + 2))$



Associative Property of Addition

$$(a + b) + c = a + (b + c)$$

Associative Property of Multiplication

$$(a \times b) \times c = a \times (b \times c)$$



Students often confuse commutative and associative properties. Consider prompting students to turn-and-talk with a partner about the difference between commutative and associative properties. Listen as they discuss, monitoring for any misconceptions that may need to be clarified with the group before closing out this component.

13.3.4 Wrap-Up: Ticket Out the Door

Component Overview: Students demonstrate their mastery of the learning goal by applying the associative, commutative, and distributive properties to generate equivalent expressions.



13.3.4 Wrap-Up: Ticket Out the Door

- Write an equivalent expression that uses each given property to complete the table.

Answers will vary.

Expression	Property	Equivalent Expression
$-2(4 + (x + -7))$	Commutative Property	$-2(4 + (-7 + x))$
$-2(4 + (x + -7))$	Associative Property	$-2((4 + x) + -7)$
$-2(4 + x + -7)$	Distributive Property	$-8 + (-2x) + 14$



To challenge students who finish early, consider having them create additional equivalent expressions using the properties. This can serve as a primer for the next lesson in which students will focus on identifying and generating equivalent expressions.